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United States Patent Application Publication

20250283576

Kind Code

A1

Publication Date

September 11, 2025

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LUBRICATION SYSTEM

Abstract

A lubrication system includes: a first pump capable of supplying oil drawn from the tank to the lubrication target; a second pump capable of discharging the oil drawn from the lubrication target to the tank; a relief valve capable of releasing the oil in the first flow path to the tank; a third pump capable of discharging the oil drawn through the intake opening to the tank, a switching valve capable of selectively switching a connection destination of the intake opening to either the lubrication target or the tank; and a valve control unit for switching the connection destination of the intake opening by controlling the switching valve.

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Family ID: 1000008518154

Appl. No.: 19/066671

Filed: February 28, 2025

Foreign Application Priority Data

JP 2024-033531

Mar. 06, 2024

Publication Classification

Int. Cl.: F16N29/02 (20060101); B64D33/00 (20060101); F16N7/40 (20060101)

U.S. Cl.:

CPC F16N29/02 (20130101); F16N7/40 (20130101); B64D33/00 (20130101); F16N2200/10 (20130101); F16N2210/08 (20130101); F16N2250/04 (20130101); F16N2250/08 (20130101); F16N2270/70 (20130101); F16N2280/00 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-033531 filed on Mar. 6, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a lubrication system.

Description of the Related Art

[0003] In recent years, in order to ensure that more people can have access to affordable, reliable, sustainable, and advanced energy, research and development is being carried out in relation to electrification technologies that contribute to energy efficiency.

[0004] JP H11-153013 A discloses a lubricating device for a vehicular internal combustion engine. The lubricating device takes up lubrication oil from an oil pan by a pump and supplies the oil to a part that needs to be lubricated (bearings of a crankshaft of the internal combustion engine, and the like).

SUMMARY OF THE INVENTION

[0005] It is desired to smoothly supply oil to a target object, such as an electric machine, to be lubricated by oil.

[0006] The present invention has the object of solving the aforementioned problem.

[0007] An aspect of the present disclosure is to provide a lubrication system including: a first pump configured to draw oil from a tank and supply the oil to a lubrication target to be lubricated by the oil; a second pump configured to draw the oil from the lubrication target and discharge the oil to the tank; a relief valve configured to release the oil to the tank from a first flow path between the first pump and the lubrication target via a second flow path different from the first flow path in a case that a pressure value of the oil in the first flow path becomes equal to or higher than a predetermined value; a third pump configured to discharge the oil drawn via an intake opening to the tank; a switching valve configured to selectively switch a connection destination of the intake opening between the lubrication target and the tank; and a valve control unit configured to control the switching valve to switch the connection destination of the intake opening of the third pump.

[0008] According to the present invention, oil can be smoothly supplied to a target object to be lubricated by the oil.

[0009] The above and other objects, features and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic view of a flying object;

[0011] FIG. 2 is a schematic diagram of a lubrication system;

[0012] FIG. 3 is a flowchart of a circuit determination process;

[0013] FIG. 4 is a diagram showing an upper limit temperature of oil that can be recovered by a second pump;

[0014] FIG. 5A is a timing chart showing the rotational speed of each pump (rotational speed of the gas turbine engine);

[0015] FIG. 5B is a timing chart showing the temperature of oil detected by a temperature sensor;
[0016] FIG. 5C is a timing chart showing the pressure of oil detected by a pressures sensor;
[0017] FIG. 5D is a timing chart showing the flow rate of oil supplied to a generator and the flow rate of oil flowing through a relief valve;
[0018] FIG. 5E is a timing chart showing the opening degree of the relief valve; and
[0019] FIG. 5F is a timing chart showing whether the lubrication circuit functions as a temperature raising circuit or a normal circuit.

DETAILED DESCRIPTION OF THE INVENTION

[0020] The lubrication system lubricates a lubrication target with oil supplied thereto. The lubrication system may be provided in a various devices. In this specification, a lubrication system mounted on a flying object will be described.

[0021] Generally, at the start of operation of the lubrication system, oil is low in temperature and high in viscosity. Such oil is less likely to flow. However, the lubrication system to be described in the present specification can quickly raise the temperature of oil after the start of operation, and the oil is likely to start to flow soon.

1. Flying Object 10

[0022] FIG. 1 is a schematic view of a flying object 10. The flying object 10 is an electric vertical take-off and landing (eVTOL) aircraft. The flying object 10 is equipped with eight VTOL rotors 12. The VTOL rotors 12 generate an upwardly directed thrust with respect to the airframe 14. The flying object 10 is equipped with eight electric motors 16. One of the electric motors 16 drives one of the VTOL rotors 12. The flying object 10 includes two cruise rotors 18. The cruise rotors 18 generate a forwardly directed thrust with respect to the airframe 14. The flying object 10 is equipped with four electric motors 20. Two of the electric motors 20 drive one of the cruise rotors 18.

[0023] The flying object 10 is equipped with a gas turbine engine (internal combustion engine) 22 and a generator 24. The gas turbine engine 22 drives the generator 24. The generator 24 generates electrical power. The electrical power generated by the generator 24 is supplied to load devices (electric motors 16, the electric motors 20, and the like) and to a power storage device (not shown) provided in the flying object 10. The gas turbine engine 22 and the generator 24 are included in a lubrication system 30 shown in FIG. 2.

2. Configuration of Lubrication System 30

[0024] FIG. 2 is a schematic diagram of a lubrication system 30. In the lubrication system 30 of the present embodiment, the target object to be lubricated by oil is the generator 24 mounted on the flying object 10. The target object to be lubricated by oil is not limited to the generator 24. The lubrication system 30 includes a lubrication circuit 32 and a control device 34.

[0025] The lubrication circuit 32 includes the generator 24 and a tank 36. The generator 24 comprises an oil inlet 24i and an oil outlet 24o. The oil flows into the generator 24 from the oil inlet 24i and flows out of the generator 24 from the oil outlet 24o. The oil circulates between the tank 36 and the generator 24.

[0026] The lubrication circuit 32 also includes a first pump 38, a flow path 40, and another flow path (first flow path) 42. The first pump 38 includes an intake opening 38i and a discharge opening 38d. The intake opening 38i is connected to the tank 36 via the flow path 40. The discharge opening 38d is connected to the oil inlet 24i of the generator 24 via the flow path 42. The first pump 38 can draw oil from the tank 36 via the flow path 40 and supply the oil to the generator 24 via the flow path 42. The first pump 38 is a feed pump 44.

[0027] The lubrication circuit 32 includes a second pump 46, a flow path 48, and another flow path 50. The second pump 46 includes an intake opening 46i and a discharge opening 46d. The intake opening 46i is connected to the oil outlet 24o of the generator 24 via the flow path 48. The discharge opening 46d is connected to the tank 36 via the flow path 50. The second pump 46 can recover (draw) oil from the generator 24 via the flow path 48 and discharge the oil to the tank 36

via the flow path **50**. The second pump **46** is included in a scavenging pump group **52**.

[0028] The lubrication circuit **32** includes a relief valve **54**, a flow path **56**, and another flow path (second flow path) **58**. The relief valve **54** includes an inlet **54i** and an outlet **54o**. The inlet **54i** is connected to the flow path **42** via the flow path **56**. The outlet **54o** is connected to the tank **36** via the flow path **58**. The outlet **54o** may be connected to the flow path **40** via the flow path **58**. The relief valve **54** opens when the pressure value of the oil on the primary side (in the flow path **56** and the flow path **42**) becomes equal to or greater than a predetermined valve-opening pressure value (predetermined value). The relief valve **54** communicates the flow path **56** and the flow path **58** (the tank **36**) in the open state. The relief valve **54** can release the oil in the flow path **42** to the tank **36** via the flow path **58**.

[0029] The lubrication circuit **32** includes a switching valve **60**, one or more third pumps **62**, and flow paths **64**, **66**, **68**, **70**. The switching valve **60** includes a first opening **60a**, a second opening **60b**, and a third opening **60c**. The switching valve **60** is a solenoid valve and is a two-position, directional control valve. While the switching valve **60** is in a non-energized (demagnetized) state, the first opening **60a** and the second opening **60b** communicate with each other. While the switching valve **60** is in an energized (excited) state, the third opening **60c** and the second opening **60b** communicate with each other. The first opening **60a** is connected to the flow path **48** via the flow path **64**. The second opening **60b** is connected to an intake opening **62i** of the third pumps **62** via the flow path **66**. The third opening **60c** is connected to the tank **36** via the flow path **70**. The switching valve **60** can selectively switch the connection destination of the intake opening **62i** of the one or more third pumps **62** between the generator **24** and the tank **36**.

[0030] The third opening **60c** and the second opening **60b** may be in communication with each other while the switching valve **60** is in the non-energized (demagnetized) state, and the first opening **60a** and the second opening **60b** may be in communication with each other while the switching valve **60** is in the energized (excited) state. Further, as the switching valve **60**, another valve (for example, a three-way valve) may be used instead of the two-position, directional control valve.

[0031] The one or more third pumps **62** include an intake opening **62i** and a discharge opening **62d**. The intake opening **62i** is connected to the second opening **60b** of the switching valve **60** via the flow path **66**. The discharge opening **62d** is connected to the tank **36** via the flow path **68**. The third pump **62** can discharge the oil drawn through the intake opening **62i** to the tank **36** through the flow path **68** and the flow path **50**. The one or more third pumps **62** are included in the scavenging pump group **52**, similarly to the second pump **46**.

[0032] As shown in FIG. 2, the lubrication circuit **32** of the present embodiment is provided with two third pumps **62**, but may be provided with one third pump **62** or three or more third pumps **62**.

[0033] The oil flowing out of the generator **24** contains gas. Therefore, the capacity of the scavenging pump group **52** that recovers oil from the generator **24** is preferably larger than the capacity of the feed pump **44** that supplies oil to the generator **24**. The lubrication circuit **32** of the present embodiment includes a plurality of scavenging pumps in the scavenging pump group **52** (the second pump **46** and one or more third pumps **62**). Therefore, the capacity of each scavenging pump in the scavenging pump group **52** may not be larger than the capacity of the feed pump **44** (first pump **38**). For example, by configuring the first pump **38**, the second pump **46**, and the third pump **62** with the same type of pump, it is possible to suppress the manufacturing cost and the maintenance cost of the lubrication system **30**.

[0034] The rotation shaft of each of the first pump **38**, the second pump **46**, and the one or more third pumps **62** is connected to the rotation shaft of the gas turbine engine **22** via a gearbox **26**. Driving the first pump **38**, the second pump **46**, and the one or more third pumps **62** by the same shaft as in the present embodiment can make the lubrication circuit **32** compact.

[0035] The lubrication circuit **32** includes a pressure sensor (pressure detection unit) **72** and a temperature sensor (temperature detection unit) **74**. The pressure sensor **72** detects a pressure value

of the oil in the flow path **42**. The temperature sensor **74** detects the temperature of the oil in the flow path **42**.

[0036] The control device **34** includes a computation unit **76** and a storage unit **78**. The control device **34** is configured, for example, as an ECU (Electronic Control Unit).

[0037] The computation unit **76** can be constituted by a processor such as a CPU (Central Processing Unit), a GPU (Graphics Processing Unit), or the like. More specifically, the computation unit **76** can be configured by a processing circuit (processing circuitry). At least a portion of the computation unit **76** may be realized by an integrated circuit such as an ASIC (Application Specific Integrated Circuit), an FPGA (Field-Programmable Gate Array), or the like. Further, at least a portion of the computation unit **76** may be constituted by an electronic circuit including a discrete device.

[0038] The computation unit **76** includes a pump control unit **80** and a valve control unit **82**. The pump control unit **80** and the valve control unit **82** can be realized by the computation unit **76** executing programs which are stored in the storage unit **78**. The pump control unit **80** controls the rotational speed of each pump. For example, the pump control unit **80** may change the gear ratio of the gearbox **26** or may control the gas turbine engine **22**. The valve control unit **82** controls the switching valve **60** based on the pressure detected by the pressure sensor **72** and the temperature detected by the temperature sensor **74**. For example, the valve control unit **82** switches the connection destination of the intake opening **62i** of the one or more third pumps **62** by switching the switching valve **60** between the energized state and the non-energized state.

[0039] The storage unit **78** is a computer-readable storage medium. The storage unit **78** is formed of a non-illustrated volatile memory and a non-illustrated nonvolatile memory. Examples of the volatile memory include, for example, a RAM (Random Access Memory) or the like. As the non-volatile memory, there may be cited, for example, a ROM (Read Only Memory), a flash memory, or the like. Data and the like are stored, for example, in the volatile memory. A program, a table, a map and the like are stored, for example, in the non-volatile memory. At least a portion of the storage unit **78** may be provided in the above-described processor, the integrated circuit, or the like.

3. Operation of Lubrication System **30**

[0040] The lubrication circuit **32** supplies oil to the generator **24** and recovers oil from the generator **24** to lubricate the generator **24**. The operation of the lubrication circuit **32** to lubricate the generator **24** is referred to herein as a lubrication operation. The lubrication operation is performed while the generator **24** is rotating (while the gas turbine engine **22** is rotating).

[0041] At the beginning of the lubrication operation, the oil is low in temperature and high in viscosity. In the case where the viscosity of the oil is high, the pressure loss of the oil in the generator **24** increases, and the pressure in the flow path **42** increases. Even if the rotational speed of each pump is increased in this state, the oil does not smoothly flow through the lubrication circuit **32**. Therefore, the lubrication system **30** performs an operation of raising the temperature of the oil while driving each pump at a rotational speed lower than a normal rotational speed until the temperature of the oil exceeds a predetermined threshold temperature. In this specification, this operation is referred to as a temperature raising operation. The lubrication system **30** performs an operation of supplying sufficient oil to the generator **24** by rotating each pump at a normal rotational speed after the temperature of the oil exceeds a predetermined threshold temperature. In this specification, this operation is referred to as normal operation. In the present specification, the predetermined threshold temperature, which is the temperature at the time of switching from the temperature raising operation to the normal operation, is referred to as a lower limit temperature for the normal operation.

[0042] When the pressure value in the flow path **42** becomes equal to or higher than the valve opening pressure value of the relief valve **54**, the relief valve **54** opens. In this manner, at the beginning of the lubrication operation, a part of the oil discharged to the flow path **42** from the first pump **38** flows through the flow path **56**, the relief valve **54**, and the flow path **58**, and returns to

the tank **36**. Therefore, the amount of oil supplied to the generator **24** is small, and the amount of oil flowing out from the generator **24** is also small.

[0043] In this state, the flow rate of the oil flowing out from the generator **24** is lower than the upper limit of the flow rate of the oil that can be recovered by the second pump **46**. In this state, the second pump **46** can recover the oil flowing out from the generator **24**. On the other hand, as long as the second pump **46** is operating in this state, the third pump **62** is not required for recovery of the oil flowing out from the generator **24**.

[0044] While a pump is in operation, the energy that the pump receives to operate is partially converted to heat. Therefore, the oil drawn by the pump is heated by the pump. The temperature of the oil is increased by continuously drawing and discharging the oil by the pump. In the lubrication circuit **32**, the pump functions as a heat source. By supplying a large amount of oil to the pump, the temperature of the oil can be quickly raised. However, in the case where the amount of oil flowing out from the generator **24** is small, only a small amount of oil is supplied to the scavenging pump group **52** (the second pump **46** and the one or more third pumps **62**).

[0045] Therefore, in the present embodiment, in the case that the amount of oil flowing out from the generator **24** is small, the third pump **62** is not used as a pump for recovering oil from the generator **24**, but is used as a pump dedicated to raising the temperature of the oil in the tank **36**. In this case, the valve control unit **82** controls the switching valve **60** to cause the third opening **60c** and the second opening **60b** to communicate with each other. Then, the intake opening **62i** of the one or more third pumps **62** communicates with the tank **36** via the flow path **66**, the switching valve **60**, and the flow path **70**. Thus, the third pump **62** draws oil from the tank **36** via the flow path **66**, the switching valve **60**, and the flow path **70**. The oil drawn into the one or more third pumps **62** is heated by the one or more third pumps **62** and the temperature of the oil rises.

[0046] As the temperature of the oil rises, the flow rate of the oil flowing through the generator **24** increases. In the case where the temperature of the oil rises to some extent, the flow rate of the oil flowing out from the generator **24** exceeds the flow rate of the oil that can be recovered by the second pump **46**. At this time, the valve control unit **82** controls the switching valve **60** to communicate the first opening **60a** and the second opening **60b** with each other. Then, the intake opening **62i** of the one or more third pumps **62** communicates with the oil outlet **240** of the generator **24** via the flow path **66**, the switching valve **60**, the flow path **64**, and the flow path **48**. Thus, the one or more third pumps **62** recover (draw) oil from the generator **24** via the flow path **66**, the switching valve **60**, the flow path **64**, and the flow path **48**. The oil recovered by the one or more third pumps **62** is heated and increased in temperature by the one or more third pumps **62**.

[0047] As described above, the valve control unit **82** controls the switching valve **60** so as to effectively utilize the scavenging pump group **52** (the second pump **46** and the one or more third pumps **62**) in accordance with the state of the lubrication circuit **32**. The valve control unit **82** controls the switching valve **60** so that the third opening **60c** and the second opening **60b** communicate with each other during a certain period at the beginning of the lubrication operation. Thus, the flow path **70** and the flow path **66** communicate with each other. In the present specification, the lubrication circuit **32** in which the flow path **70** and the flow path **66** communicate with each other is referred to as a temperature raising circuit. The valve control unit **82** controls the switching valve **60** such that the first opening **60a** and the second opening **60b** communicate with each other after the certain period described above ends. Thus, the flow path **64** and the flow path **66** communicate with each other. In the present specification, the lubrication circuit **32** in which the flow path **64** and the flow path **66** communicate with each other is referred to as a normal circuit.

4. Circuit Determination Process

[0048] FIG. **3** is a flowchart of a circuit determination process. The circuit determination process is a process in which the valve control unit **82** selects one of the temperature raising circuit and the normal circuit. When the gas turbine engine **22** is started, the first pump **38**, the second pump **46**,

and the one or more third pumps **62** are also started. Thus, the lubrication operation is started. During the lubrication operation, the valve control unit **82** performs a circuit determination process described below at predetermined time intervals.

[0049] In step **S1**, the valve control unit **82** determines whether or not the relief valve **54** is opened. The valve control unit **82** determines that the relief valve **54** is opened when the pressure value detected by the pressure sensor **72** is equal to or higher than the valve opening pressure value of the relief valve **54**. On the other hand, the valve control unit **82** determines that the relief valve **54** is not opened when the pressure value detected by the pressure sensor **72** is less than the valve opening pressure value of the relief valve **54**. The valve opening pressure value of the relief valve **54** is stored in the storage unit **78** in advance. In the case that the relief valve **54** is opened (step **S1**: YES), the process transitions to step **S2**. On the other hand, in the case that the relief valve **54** is not opened (step **S1**: NO), the process proceeds to step **S4**.

[0050] Upon transitioning from step **S1** to step **S2**, the valve control unit **82** determines whether the oil flowing out from the generator **24** can be recovered only by the second pump **46**. Specifically, if the flow rate of the oil flowing out from the generator **24** is equal to or less than the flow rate of the oil recoverable only by the second pump **46**, then the valve control unit **82** determines that all the oil flowing out from the generator **24** can be recovered only by the second pump **46**. On the other hand, the valve control unit **82** determines that the recovery of all the oil flowing out of the generator **24** cannot be covered by the second pump **46** alone if the flow rate of the oil flowing out of the generator **24** exceeds the maximum possible flow rate of the oil that can be recovered by the second pump **46**. The determination in step **S2** will be described below.

[0051] In a state where the relief valve **54** is opened, the pressure value of the oil flowing through the lubrication circuit **32** is a constant value (valve opening pressure value of the relief valve **54**). In this state, the flow rate of the oil flowing out from the generator **24** depends on the temperature of the oil. On the other hand, the flow rate of the oil that can be recovered by the second pump **46** depends on the rotational speed of the second pump **46**. Here, the characteristics shown in FIG. **4** are established by replacing the flow rate of the oil flowing out from the generator **24** with the temperature of the oil and replacing the maximum possible flow rate of the oil that is recoverable by the second pump **46** with the rotational speed of the second pump **46**.

[0052] The characteristics shown in FIG. **4** indicate the upper limit temperature of the oil that can be recovered by the second pump **46**. The upper limit temperature correlates with the maximum possible flow rate of the oil that is recoverable by the second pump **46**. As shown in FIG. **4**, the upper limit temperature increases as the rotational speed increases. The characteristics shown in FIG. **4** can be obtained by actually performing the lubrication operation. For example, the characteristics shown in FIG. **4** can be acquired by actually measuring the flow rate of the oil that can be recovered by the second pump **46** and the temperature of the oil for each rotational speed of the second pump **46**. The characteristics shown in FIG. **4** are stored in the storage unit **78** in advance. The valve control unit **82** determines whether or not the oil flowing out from the generator **24** can be recovered only by the second pump **46** based on the characteristics and the temperature detected by the temperature sensor **74**.

[0053] In the case that the temperature detected by the temperature sensor **74** is equal to or lower than the upper limit temperature determined according to the rotational speed of the second pump **46**, the valve control unit **82** determines that the oil flowing out of the generator **24** can be recovered only by the second pump **46**. On the other hand, in the case that the temperature detected by the temperature sensor **74** exceeds the upper limit temperature determined in accordance with the rotational speed of the second pump **46**, the valve control unit **82** determines that the oil flowing out from the generator **24** is too much to be recovered only by the second pump **46**.

[0054] In the case that the oil flowing out of the generator **24** can be recovered only by the second pump **46** (step **S2**: YES), the process transitions to step **S3**. On the other hand, in the case that the oil flowing out of the generator **24** beyond the amount that can be recovered only by the second

pump **46** (step S2: NO), the process transitions to step S4.

[0055] Upon transitioning from step S2 to step S3, the valve control unit **82** selects the temperature raising circuit. The valve control unit **82** controls a valve drive circuit (not shown) so that the switching valve **60** is in an energized (excited) state. Then, the switching valve **60** is put in an energized (excited) state, and the third opening **60c** and the second opening **60b** communicate with each other. The intake opening **62i** of the one or more third pumps **62** communicates with the tank **36**. As a result, the lubrication circuit **32** functions as a temperature raising circuit.

[0056] In the temperature raising circuit, the second pump **46** recovers (draws) the oil flowing out of the generator **24** via the flow path **48**, and discharges the recovered oil to the tank **36** via the flow path **50**. On the other hand, the third pump **62** draws oil from the tank **36** via the flow path **70**, the switching valve **60**, and the flow path **66**, and discharges the drawn oil to the tank **36** via the flow path **68** and the flow path **50**.

[0057] Upon transitioning from to step S1 or from step S2 to step S4, the valve control unit **82** selects the normal circuit. The valve control unit **82** controls a valve drive circuit (not shown) so that the switching valve **60** is put in a non-energized (demagnetized) state. Then, the switching valve **60** is de-energized (demagnetized), and the first opening **60a** and the second opening **60b** communicate with each other. The intake opening **62i** of the one or more third pumps **62** communicates with the oil outlet **240** of the generator **24**. As a result, the lubrication circuit **32** functions as a normal circuit.

[0058] In the normal circuit, the second pump **46** recovers (draws) the oil flowing out from the generator **24** via the flow path **48**, and discharges the recovered oil to the tank **36** via the flow path **50**. On the other hand, the one or more third pumps **62** also recover (draw) the oil flowing out from the generator **24** via the flow path **48**, the flow path **64**, the switching valve **60**, and the flow path **66**, and discharge the recovered oil to the tank **36** via the flow path **68** and the flow path **50**.

5. Timing Chart of Lubrication Operation

[0059] FIG. 5A is a timing chart showing the rotational speed of each pump (rotational speed of the gas turbine engine **22**). FIG. 5B is a timing chart showing the temperature of oil detected by a temperature sensor **74**. FIG. 5C is a timing chart showing the pressure of oil detected by a pressures sensor **72**. FIG. 5D is a timing chart showing the flow rate of oil supplied to the generator **24** and the flow rate of oil flowing through the relief valve **54**. FIG. 5E is a timing chart showing the opening degree of the relief valve **54**. FIG. 5F is a timing chart showing whether the lubrication circuit **32** functions as a temperature raising circuit or a normal circuit.

[0060] At time T1, the gas turbine engine **22** is turned on. As shown in FIG. 5A, the pump control unit **80** starts the first pump **38**, the second pump **46**, and the one or more third pumps **62** in the low rotational speed range. Thus, the temperature raising operation is started. Then, as shown in FIG. 5C, the pressure in the flow path **42** is rapidly increased. As a result, the relief valve **54** opens as shown in FIG. 5E, and the oil starts to flow through the relief valve **54** as shown in FIG. 5D. As shown in FIG. 5F, the valve control unit **82** causes the lubrication circuit **32** to function as the temperature raising circuit. That is, the valve control unit **82** causes the third opening **60c** and the second opening **60b** of the switching valve **60** to communicate with each other.

[0061] As shown in FIG. 5B, the temperature of the oil gradually rises as the oil circulates through the temperature raising circuit. As the temperature of the oil rises, the viscosity of the oil gradually decreases, and the pressure loss in the generator **24** gradually decreases. Then, as shown in FIG. 5E, the opening degree of the relief valve **54** gradually narrows. As a result, as shown in FIG. 5D, the flow rate of the oil supplied to the generator **24** gradually increases, and the flow rate of the oil flowing through the relief valve **54** gradually decreases.

[0062] As shown in FIG. 5D, at time T2, the flow rate of the oil supplied to the generator **24** exceeds the maximum flow rate of the oil that can be recovered only by the second pump **46**. As shown in FIG. 5B, the temperature of the oil exceeds the upper limit temperature of the oil corresponding to the maximum possible flow rate of the oil that is recoverable by the second pump

46. As shown in FIG. 5F, at this timing, the valve control unit **82** causes the lubrication circuit **32** to function as a normal circuit. That is, the valve control unit **82** causes the first opening **60a** and the second opening **60b** of the switching valve **60** to communicate with each other.

[0063] As shown in FIG. 5E, the relief valve **54** is closed at time T3. In this state, the oil circulates in the normal circuit. Thus, the temperature of the oil continues to rise as shown in FIG. 5B. At time T4, the temperature of the oil reaches the normal operation lower limit temperature. As shown in FIG. 5A, at this timing, the pump control unit **80** operates the first pump **38**, the second pump **46**, and the third pump **62** in the normal rotational speed range higher than the low rotational speed range. Thus, the normal operation is started.

6. Others

[0064] The lubrication system **30** may include an electric motor instead of the gas turbine engine **22**.

7. Effects of Present Embodiment

[0065] In the present embodiment, the connection destination of the intake opening **62i** of the one or more third pumps **62** is selectively switched to either the generator **24** or the tank **36**. According to the present embodiment, the one or more third pumps **62** can be used not only as a pump for recovering oil from the generator **24**, but also as a pump dedicated to heating oil. According to the present embodiment, the temperature of the oil can be quickly raised by switching the switching valve **60** in accordance with the flow state of the oil. Thus, oil is likely to flow at an early stage, and thus oil can be smoothly supplied to the target object to be lubricated by the oil.

8. Supplementary Note

[0066] In relation to the above-described disclosure, the following supplementary notes are further disclosed.

Supplementary Note 1

[0067] The lubrication system (**30**) according to the present disclosure including: the first pump (**38**) configured to draw oil from the tank (**36**) and supply the oil to the lubrication target (**24**) to be lubricated by the oil; the second pump (**46**) configured to draw the oil from the lubrication target and discharge the oil to the tank; the relief valve (**54**) configured to release the oil to the tank from the first flow path (**42**) located between the first pump and the lubrication target via the second flow path (**58**) different from the first flow path in the case that a pressure value of the oil in the first flow path becomes equal to or higher than the predetermined value; the third pump (**62**) configured to discharge the oil drawn via the intake opening (**62i**) to the tank; the switching valve (**60**) configured to selectively switch the connection destination of the intake opening between the lubrication target and the tank; and the valve control unit (**82**) configured to control the switching valve to switch the connection destination of the intake opening of the third pump.

[0068] In the above configuration, the connection destination of the intake opening of the third pump can be selectively switched to either the lubrication target or the tank. According to the above configuration, the third pump can be used not only as a pump for recovering oil from the lubrication target, but also as a pump dedicated to heating of oil. According to such a configuration, the temperature of the oil can be quickly raised by switching the switching valve in accordance with the flow state of the oil. Thus, the oil is likely to flow at an early stage, and thus the oil can be smoothly supplied to the target object to be lubricated by the oil.

Supplementary Note 2

[0069] In the lubrication system according to Supplementary Note 1, the valve control unit may perform first control of the switching valve to connect the intake opening of the third pump to the tank while the oil is released from the first flow path to the tank via the relief valve and the second flow path.

Supplementary Note 3

[0070] In the lubrication system according to Supplementary Note 2, the valve control unit may switch control of the switching valve from the first control to second control to connect the intake

opening of the third pump to the lubrication target in a case that the flow rate of the oil discharged from the lubrication target exceeds the flow rate of the oil drawable by the second pump.

Supplementary Note 4

[0071] The lubrication system according to any one of Supplementary Notes 1 to 3, further comprising: the pressure detection unit (72) configured to detect the pressure value of the oil in the first flow path; and the temperature detection unit (74) configured to detect a temperature of the oil in the first flow path, wherein the valve control unit may control the switching valve based on the pressure value detected by the pressure detection unit and the temperature detected by the temperature detection unit.

[0072] The flow rate of the oil depends on the temperature of the oil. Therefore, the control related to the flow rate of the oil can be performed by detecting the temperature of the oil instead of detecting the flow rate of the oil. Cost can be reduced by using a temperature sensing device instead of a flow rate sensing device.

Supplementary Note 5

[0073] In the lubrication system according to Supplementary Note 4, the valve control unit may control the switching valve to connect the intake opening of the third pump to the tank in the case that the pressure value detected by the pressure detection unit is equal to or higher than the predetermined value and the temperature detected by the temperature detection unit is equal to or lower than an upper limit temperature determined correspondingly to a rotational speed of the second pump.

Supplementary Note 6

[0074] In the lubrication system according to any one of Supplementary Notes 1 to 5, the first pump may be the feed pump (44), and the second pump and the third pump may be scavenging pumps (52).

[0075] According to the above configuration, since the plurality of pumps, that is, the second pump and the third pump, are made to function as scavenging pumps, it is not necessary to make each of the capacity of the second pump and the capacity of the third pump larger than the capacity of the first pump. Therefore, the first pump, the second pump, and the third pump can be configured by pumps of the same type. This makes it possible to reduce the manufacturing cost and the maintenance cost of the lubrication system.

Supplementary Note 7

[0076] In the lubrication system according to any one of Supplementary Notes 1 to 6, the first pump, the second pump, and the third pump may be driven by the common actuator.

[0077] According to the above configuration, the first pump, the second pump, and the third pump are driven by the common actuator, and thus the circuit can be made compact.

Supplementary Note 8

[0078] In the lubrication system according to Supplementary Note 7, the actuator may be the internal combustion engine (22).

Supplementary Note 9

[0079] In the lubrication system according to Supplementary Note 7, the actuator may be the electric motor.

[0080] Although concerning the present disclosure, a detailed description thereof has been presented above, the present disclosure is not necessarily limited to the individual embodiments described above. These embodiments may be subjected to various additions, substitutions, modifications, partial deletions and the like, within a range that does not deviate from the essence and gist of the present disclosure, or the spirit of the present disclosure as derived from the contents described in the claims and equivalents thereof. Further, the embodiments can also be implemented together in combination. For example, in the above-described embodiments, the order of each of the operations and the order of each of the processes are illustrated as examples, and the present invention is not necessarily limited to these features. The same also applies to cases in which

numerical values or mathematical expressions are used in the description of the aforementioned embodiments.

Claims

1. A lubrication system comprising: a first pump configured to draw oil from a tank and supply the oil to a lubrication target to be lubricated by the oil; a second pump configured to draw the oil from the lubrication target and discharge the oil to the tank; a relief valve configured to release the oil to the tank from a first flow path located between the first pump and the lubrication target via a second flow path different from the first flow path in a case that a pressure value of the oil in the first flow path becomes equal to or higher than a predetermined value; a third pump configured to discharge the oil drawn via an intake opening to the tank; a switching valve configured to selectively switch a connection destination of the intake opening between the lubrication target and the tank; and a valve control unit configured to control the switching valve to switch the connection destination of the intake opening of the third pump.
 2. The lubrication system according to claim 1, wherein the valve control unit performs first control of the switching valve to connect the intake opening of the third pump to the tank while the oil is released from the first flow path to the tank via the relief valve and the second flow path.
 3. The lubrication system according to claim 2, wherein the valve control unit switches control of the switching valve from the first control to second control to connect the intake opening of the third pump to the lubrication target in a case that a flow rate of the oil discharged from the lubrication target exceeds a flow rate of the oil drawable by the second pump.
 4. The lubrication system according to claim 1, further comprising: a pressure detection unit configured to detect the pressure value of the oil in the first flow path; and a temperature detection unit configured to detect a temperature of the oil in the first flow path, wherein the valve control unit controls the switching valve based on the pressure value detected by the pressure detection unit and the temperature detected by the temperature detection unit.
 5. The lubrication system according to claim 4, wherein the valve control unit controls the switching valve to connect the intake opening of the third pump to the tank in a case that the pressure value detected by the pressure detection unit is equal to or higher than the predetermined value and the temperature detected by the temperature detection unit is equal to or lower than an upper limit temperature determined correspondingly to a rotational speed of the second pump.
 6. The lubrication system according to claim 1, wherein the first pump is a feed pump, and the second pump and the third pump are scavenging pumps.
 7. The lubrication system according to claim 1, wherein the first pump, the second pump, and the third pump are driven by a common actuator.
 8. The lubrication system according to claim 7, wherein the actuator is an internal combustion engine.
 9. The lubrication system according to claim 7, wherein the actuator is an electric motor.
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